JOHANNES **KRUSE**, Ph.D.

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Summary: I hold a Ph.D. in computer science with a specialization in Artificial Intelligence (AI), focusing on machine learning (ML), natural language processing (NLP), and recommender systems (RS). My expertise spans sequential modeling using transformers, recurrent neural networks (RNNs), and convolutional neural networks (CNNs), alongside theoretical work in reinforcement learning (RL) and multi-armed bandits. I have applied these techniques to develop and deploy large-scale AI solutions that serve millions of users worldwide.

PUBLICATIONS & PREPRINTS (11)

- **Johannes Kruse**, Ryotaro Shimizu, Kasper Lindskow, Jon Tofteskov, Michael Riis Andersen, Julian McAuley, and Jes Frellsen. 2026. *ZoRRO: A Zero-Weight Personalized Recommender System for Scalable News Recommendation.* In Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR '26**). https://doi.org/10.1145/3805712.3809909
- Árni Már Einarsson, Elisabetta Petrucci, Jannie Møller Hartley, Stine Lomborg, and **Johannes Kruse**. 2025. *"I Must Have Clicked on Something" – Users' Experiences and Evaluations of News Recommender Systems*. **Journalism Practice**. https://doi.org/10.1080/17512786.2025.2572972
- **Johannes Kruse**, Kasper Lindskow, Michael Riis Andersen, Ryotaro Shimizu, Julian McAuley, Pierre-Alexandre Mattei, and Jes Frellsen. 2025. *Normative Alignment of Recommender Systems via Internal Label Shift.* In Proceedings of the Nineteenth ACM Conference on Recommender Systems (**RecSys '25**). https://doi.org/10.1145/3705328.3759309
- **Johannes Kruse**, Kasper Lindskow, Michael Riis Andersen, and Jes Frellsen. 2025. *Why design choices matter in recommender systems*. **Nature Machine Intelligence**. https://doi.org/10.1038/s42256-025-01043-5
- Ryotaro Shimizu, Takashi Wada, Yu Wang, **Johannes Kruse**, Sean O'Brien, Sai HtaungKham, Linxin Song, Yuya Yoshikawa, Yuki Saito, Fugee Tsung, Masayuki Goto, and Julian McAuley. 2025. *Disentangling Likes and Dislikes in Personalized Generative Explainable Recommendation*. In Proceedings of the ACM on Web Conference 2025 (**WWW '25**). https://doi.org/10.1145/3696410.3714583
- **Johannes Kruse**, Kasper Lindskow, Saikishore Kalloori, Marco Polignano, Claudio Pomo, Abhishek Srivastava, Anshuk Uppal, Michael Riis Andersen, and Jes Frellsen. 2024. *RecSys Challenge 2024: Balancing Accuracy and Editorial Values in News Recommendations*. In Proceedings of the 18th ACM Conference on Recommender Systems (**RecSys '24**). https://doi.org/10.1145/3640457.3687164
- **Johannes Kruse**, Kasper Lindskow, Saikishore Kalloori, Marco Polignano, Claudio Pomo, Abhishek Srivastava, Anshuk Uppal, Michael Riis Andersen, and Jes Frellsen. 2024. *EB-NeRD a large-scale dataset for news recommendation*. In Proceedings of the Recommender Systems Challenge 2024 (**RecSysChallenge '24**). https://doi.org/10.1145/3687151.3687152
- Alain Starke, Sanne Vrijenhoek, Lien Michiels, **Johannes Kruse**, and Nava Tintarev. 2024. *Report on NORMalize: The Second Workshop on the Normative Design and Evaluation of Recommender Systems*. In Proceedings of CEUR Workshop Proceedings (**CEUR-WS.org**). https://ceur-ws.org/Vol-3898/
- **Johannes Kruse**, Kasper Lindskow, Michael Riis Andersen, and Jes Frellsen. 2023. *Creating the next generation of news experience on ekstrabladet.dk with recommender systems*. In Proceedings of the 17th ACM Conference on Recommender Systems (**RecSys '23**). https://doi.org/10.1145/3604915.3610248
- Sanne Vrijenhoek, Lien Michiels, **Johannes Kruse**, Alain Starke, Jordi Viader Guerrero, and Nava Tintarev. 2023. *Report on NORMalize: The First Workshop on the Normative Design and Evaluation of Recommender Systems*. In Proceedings of CEUR Workshop Proceedings (**CEUR-WS.org**). https://ceur-ws.org/Vol-3639/
- **Johannes Kruse** and Lars Kai Hansen. 2019. *Det Etiske Råd – Redegørelse om sundhedswearables og big data*. pp. 116–141. https://etiskraad.dk/alle-udgivelser/sundhedsdata/sundhedswearables-og-big-data

WORKSHOPS, TUTORIALS, DEMOS & OTHERS (7)

- Jon Tofteskov, **Johannes Kruse**, Sanne Vrijenhoek, Annelien Smets. 2026. *Recommenders in News Media@NAMS 2026 Side Event*. https://sites.google.com/view/recommenders-in-news-media
- Lien Michiels, Sanne Vrijenhoek, Alain Starke, **Johannes Kruse**, and Savvina Daniil. 2025. *NORMalize 2025: The Third Workshop on Normative Design and Evaluation of Recommender Systems*. In Proceedings of the 19th ACM Conference on Recommender Systems (**RecSys '25**). https://doi.org/10.1145/3705328.3748002
- **Johannes Kruse**, Kasper Lindskow, Anshuk Uppal, Michael Riis Andersen, and Jes Frellsen. 2024. *RecSys Challenge 2024: Balancing Accuracy and Editorial Values in News Recommendations*. (**RecSys '24**). https://recsys.eb.dk/
- Alain Starke, Sanne Vrijenhoek, Lien Michiels, **Johannes Kruse**, and Nava Tintarev. 2024. *NORMalize 2024: The Second Workshop on Normative Design and Evaluation of Recommender Systems*. In Proceedings of the 18th ACM Conference on Recommender Systems (**RecSys '24**). https://doi.org/10.1145/3640457.3687103
- Alan Said, Toine Bogers, Christoph Trattner, Simen Eide, Benjamin Kille, Bruce Ferwerda, and **Johannes Kruse**. 2024. *Nordic Personalization Workshop*. https://personalizationday.github.io/2024/

- **Johannes Kruse**, Lien Michiels, Alain Starke, Nava Tintarev, and Sanne Vrijenhoek. 2024. *NORMalize: A Tutorial on the Normative Design and Evaluation of Information Access Systems*. In Proceedings of the 2024 Conference on Human Information Interaction and Retrieval (**CHIIR '24**). https://doi.org/10.1145/3627508.3638319
- Sanne Vrijenhoek, Lien Michiels, **Johannes Kruse**, Alain Starke, Nava Tintarev, and Jordi Viader Guerrero. 2023. *NORMalize: The First Workshop on Normative Design and Evaluation of Recommender Systems*. In Proceedings of the 17th ACM Conference on Recommender Systems (**RecSys '23**). https://doi.org/10.1145/3604915.3608757

EXPERIENCE

04/2021 – Present **JP/Politikens Media Group** | **Applied AI Scientist** **Copenhagen, DK**
- Built and deployed the first large-scale recommender system for eb.dk, increasing subscription sales by more than 38%.
- Led the end-to-end development and management of recommendation systems team and core products.

07/2020 – 09/2020 **Infosys Limited** | **InStep Intern** **Bangalore, IND**
- Developed a scalable framework for a conversational search component integrated into an enterprise search engine.
- Built AI-driven modules for question answering and generation using state-of-the-art machine learning techniques.

01/2020 – 09/2020 **Implement Consultant Group** | **Junior Consultant** **Copenhagen, DK**
- Contributed to the Data & Analytics team by optimizing value chains through advanced machine learning solutions.
- Code contributor to multiple text mining projects, extracting and visualizing topics for document classifcation.

09/2018 – 09/2019 **AudienceProject** | **Data Analyst** **Copenhagen, DK**
- Developed Python-based software for structural sentence recognition, enhancing document categorization workflows.
- Collaborated with cross-functional teams to implement solutions tailored to the needs of major departments.

03/2018 – 09/2018 **The Danish Ethics Committee** | **External Consultant** **Copenhagen, DK**
- Developed a working paper titled "Privacy-by-Design – Teknisk Notat," exploring technical solutions such as differential privacy, k-anonymity, and multiparty computation to enable citizens to gain full control over their personal data in public spaces.
- Facilitated discussions with leading cryptography experts and summarized findings for the Ethics Committee.

06/2018 – 08/2018 **Engineering World Health (EWH)** | **Expatriate Volunteer** **Dolakha District, NP**
- Stationed as an engineer in a remote Himalayan hospital in Nepal for six weeks, providing technical support and maintenance.
- Collaborated with Nepal's Ministry of Health and local hospital staff to advance the UN's 2030 Sustainable Development Goals, ensuring access to universal health benefits.

08/2016 – 06/2018 **Technical University of Denmark** | **Teaching Assistant** **Kgs. Lyngby, DK**
- Clarified learning objectives and fostered productive discussions on Calculus and Algebra courses.
- Supported professors by evaluating and grading mathematical assignments to ensure academic standards.

02/2016 – 05/2018 **Novozymes** | **Student Assistant** **Bagsvaerd, DK**
- Calibrated and maintained laboratory equipment to ensure precise experimental results.
- Enhanced calibration workflows in the Household Care division, boosting capacity from four to six washing machines.

08/2017 – 12/2017 **Rensselaer Polytechnic Institute** | **Research Assistant** **New York, US**
- Conducted population studies for the Cognitive Science Department under Prof. Wayne Gray, following strict scientific protocols.

EDUCATION

12/2021 – 06/2025 **Technical University of Denmark** | **Ph.D. Student** **Kgs. Lyngby, DK**
- Project: Responsible Recommender Systems for Danish News Publishing (RRS-DK).
- The project was an industrial Ph.D. collaboration between JP/Politikens Media Group and the Technical University of Denmark.

03/2024 – 06/2025 **University of California San Diego** | **Research Scholar** **San Diego, CA, US**
- Conducting research in Prof. Julian McAuley's lab on innovative recommender system architectures and responsible AI.

09/2018 – 04/2021 **Technical University of Denmark** | **MSc Applied Mathematics** **Kgs. Lyngby, DK**
- Thesis: Deep learning for Natural Language Processing (NLP). Exploring the use of Knowledge Graph technologies with the objective of enriching word embeddings. Grade: 12 (A+).
- Key Courses: Machine Learning, Advanced Modeling, Time Series Analysis, and Constrained Optimization.
- Contributed to Danish research project *WriteReader* by evaluating machine-generated captions using Recurrent Neural Networks and Convolutional Neural Networks.

09/2019 – 01/2020 **Barcelona School of Informatics** | **MSc Exchange Student** **Barcelona, ES**
- Key courses: Deep Learning, Kernel-based Machine Learning and Multivariate Modelling, and Human Language Engineering.

09/2015 – 06/2018 **Technical University of Denmark** | **BSc Biomedical Engineering** **Kgs. Lyngby, DK**
- Thesis: Computational Neuroimaging. Investigation of transcranial magnetic stimulation variability in defined brain states by applying data analysis using Python programming. Grade: 12 (A+).
- Courses: Advanced Engineering Mathematics, Physics, General Chemistry, Human Biology, Medical Imaging.

08/2017 – 12/2017 **Rensselaer Polytechnic Institute** | **BSc Exchange Student** **Troy, NY, US**
- Key courses: Biomechanics, Biomedical Product Development, Skilled Performance & Training.

SUPERVISION AND MENTORSHIP IN DEEP LEARNING PROJECTS

- Deep learning Course, News Recommendation Systems, Technical University of Denmark (2024). Supervised 26 groups involving more than 100 students working on the RecSys '24 challenge as part of the deep learning course at the Technical University of Denmark, providing guidance on advanced recommender systems techniques and strategies. Role: Co-Supervisor.
- Zhijian Feng, 2023, Technical University of Denmark & Ekstra Bladet. MSc thesis: *Using Image Information in Deep Learning-Based Recommender Systems for News.* Role: Main Supervisor.
- Magnus Waldemar Hoff Harder & Nikolaj Bach Meineche, 2023, Technical University of Denmark & Ekstra Bladet. BSc thesis: *Transformer-based News Recommendations: Leveraging User Click-History with Deep Learning.* Role: Main Supervisor.

INVITED TALKS

2026	Responsible Recommender Systems	CWI - HCDA Group
2025	Open sourcing for Innovation	DN Media Group
2025	RecSys '24: Balancing Accuracy and Editorial Values in News Recommendations	Nordic AI Journalism
2024	Open-Sourcing Datasets and Designing Machine Learning Competitions	University of California San Diego
2023	Creating the Next Generation of News Experience with Recommender Systems	Nordic AI in Media Summit
2023	Creating the Next Generation of News Experience at Ekstra Bladet with AI	Nordic Personalization Day
2022	Do Recommendation Systems Generalize Across News Domains?	The University of Melbourne
2022	The Platform Intelligence in News Project	RecSys - DK

ACADEMIC SERVICE & PEER REVIEW

Special Interest Group on Information Retrieval (**SIGIR**)	2026
The Web Conference (**WWW**)	2026
The CHI conference on Human Factors in Computing Systems (**CHI**)	2026
Conference on Recommender Systems (**RecSys**)	2023, 2024, 2025
Conference on Neural Information Processing Systems (**NeurIPS**)	2025
Human Information Interaction and Retrieval (**CHIIR**)	2024

AWARDS

2024	Best Use of AI in Revenue Strategy Digital Media Worldwide
2024	Best Use of AI in Revenue Strategy Digital Media Awards Europe
2023	Data and AI-driven Transformation Project of the Year Nordic DAIR Awards
2023	Best Product Iteration Global Media Awards

These awards reflect the combined efforts and accomplishments of our talented team at Platform Intellenge in News.

SCHOLARSHIPS

2024	$26,000 in travel grants to support my research visit to the University of California San Diego.
2022	$2,000 travel grant from DDSA to attent 16th ACM Conference on Recommender Systems, Seattle, WA, USA.
2019	$11,000 in travel grants to support study abroad at Barcelona School of Informatics during master's studies.
2018	$55,000 was collectively raised by 16 people to support our six-week volunteer work for EWH in Nepal.
2017	$4,500 in travel grants to support study abroad at Rensselaer Polytechnic Institute during bachelor's studies.

$ = USD.

OTHER

Languages	Danish (native), English (fluent), Spanish (intermediate).
Coding	Python (expert), R (proficient), SQL (intermediate), HTML (familiar).
AI-tools	Codex, Claude Code, ChatGPT, GitHub Copilot.
Frameworks	PyTorch, TensorFlow, Hugging Face, Scikit-learn, SciPy, Polars, Pandas, MLFlow, Matplotlib, NumPy.
Passions	Culture, climbing, freediving, hiking, and endurance sports. Lived abroad in the United States, Australia, and Greenland, and Spain; certified rock-climbing instructor; completed a 28-meter freedive, multiple ultramarathons, and an Ironman.